

APPENDIX E — PRIVACY-PRESERVING OUTPUT FORMATS AND USER INTERFACE REPRESENTATIONS (NON-LIMITING)

A. Overview

This appendix provides non-limiting examples of privacy-preserving output formats and user interface representations that may be used to present results derived from one or more nodes. Outputs are configured to avoid explicit anatomical imagery and avoid explicit anatomical labeling while still conveying useful longitudinal patterns, context tags, and confidence indicators.

B. Output Types (Non-Limiting)

In various embodiments, the computing system generates one or more output types including: 1. Indices (single-value scores) presented on bounded scales (e.g., 0–1, 0–100, or zone-based scales). 2. Zones and bands indicating range placement (e.g., Low / Typical / Elevated) and changes over time. 3. Curves and trend lines indicating longitudinal change across days or nights. 4. Distributions indicating stability or variability across windows. 5. Context tags indicating state labels such as sleep, rest, activity, posture proxies, thermal context, and wear integrity. 6. Confidence indicators indicating validity and data quality, including motion gating results and contact integrity status.

C. Example Longitudinal Views (Non-Limiting)

In certain embodiments, the interface presents longitudinal views including: a nightly or daily timeline with segmented windows labeled by context tags, a trend dashboard showing change-over-time for selected indices, and a stability panel showing variability and repeatability across sessions. In certain embodiments, the interface provides a quality layer showing which windows were accepted, down-weighted, or excluded.

D. Example Baseline and Geometry Views (Non-Limiting)

In certain embodiments, baseline and geometry workflows generate outputs presented as abstracted references, including: a baseline reference card listing derived indices and ranges, a profile curve representing a geometry-derived or compliance-derived signature without explicit anatomical depiction, and a repeatability score indicating whether baseline acquisition was stable across repeated trials. In certain embodiments, the interface includes scan confidence indicators and session validity summaries.

E. Example Cross-Device Correlation Views (Non-Limiting)

In certain embodiments, cross-device correlation is presented as: 1. an alignment score computed across selected nodes, 2. a synchrony curve indicating matched timing behavior, 3. a lag signature indicating time offsets between changes in one node and another, and 4. a context-filter toggle restricting correlation to matched windows (e.g., low-motion sleep windows). In certain embodiments, the interface indicates which nodes contributed to a given correlation view and provides confidence labeling.

F. Example Cross-User Correlation Views (Non-Limiting)

In certain embodiments, cross-user correlation outputs are presented as privacy-preserving summaries including: an alignment score and stability indicator, a synchrony summary derived from matched context windows, and a confidence indicator reflecting data quality, recency, and consent scope. In certain

embodiments, the interface displays consent status (active, expired, revoked), share scope (which metric categories are shared), and a local-only or derived-only mode indicator.

G. Non-Limiting UI Controls and Safety Elements

In various embodiments, the interface includes one or more of: 1. a privacy mode selector (local-only, derived-only, consent-gated exchange), 2. a share scope selector for metric categories, 3. a revocation control for cross-user permissions, 4. a data deletion control for local derived metrics and/or raw caches, and 5. a misuse resistance indicator that blocks outputs when quality thresholds are not met.

H. Variations

Any output type described herein may be substituted with equivalent representations that preserve privacy while conveying longitudinal change, context conditioning, and correlation results. Output formats may be adapted for clinical-facing views, consumer-facing views, or mixed modes, while maintaining privacy-preserving presentation.